

TAM THUC STRATEGEM

Monitoring, Observability and Alerting v1.0

1. CÁC LOẠI METRICS CẦN THEO DÕI

****Business Metrics****:

- Số lượng queries/ngày
- Conversion rate (free → premium)
- Retention D1, D7, D30

****Technical Metrics****:

- Chart generation time (p50, p95, p99)
- LLM response time
- Error rate
- Database query performance

****Quality Metrics****:

- Expert validation pass rate
- User satisfaction (NPS hoặc in-app rating)

2. TOOLING ĐỀ XUẤT

- ****Metrics & Dashboards****: Prometheus + Grafana
- ****Error Tracking****: Sentry
- ****Logging****: Loki hoặc Datadog
- ****Alerting****: Alertmanager hoặc PagerDuty (cho production)
- ****User Analytics****: PostHog hoặc Mixpanel

3. ALERTING QUAN TRỌNG

- Chart generation time > 5s (p95)
- Error rate > 1%
- LLM API downtime hoặc rate limit
- Database connection issues